



ROOT CAUSE ANALYSIS

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

VCF 9 Multi-Layer Cascade Root Cause Analysis & Recovery

One expired evaluation license on the domain controller took out DNS, VCF Operations, licensing, Fleet Management, and credential management. This is how it unravelled — and the verified fix for every layer.

DC EVAL EXPIRY

DNS

VCF OPERATIONS

LICENSING

FLEET MANAGEMENT

RESOURCE LOCKS

ESXI SVC ACCOUNT

v1.0

VMware Cloud Foundation 9.0.1

VCF 9 Multi-Layer Cascade — Root Cause Analysis & Recovery

Document Information

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The whole thing in one sentence: the Active Directory domain controller runs **Windows Server 2019 Evaluation**; its 180-day eval expired, so `wlms.exe` began force-shutting it down roughly hourly — and because that DC is the environment's **only DNS server**, every VCF component that resolves a peer by name began failing in turn.

1. Executive Summary

A single, non-obvious root cause — an **expired Windows Server evaluation license on the AD/DNS domain controller** — produced a chain of failures that *looked like* six unrelated product problems:

1. **DNS outages** every time the DC auto-shut-down.
2. **VCF Operations** "Unable to connect to VC" on the vCenter and vSAN adapters.
3. **License Management** "Licenses could not be synchronized with the vCenter systems."
4. **Fleet Management** (Tasks / Passwords) "Service or view is not available at this time."
5. **4 stuck SDDC Manager resource locks** blocking all credential operations.
6. **A missing ESXi service account** on esxi04.

Only #5 and #6 were pre-existing; **#1–#4 were all the same DNS problem** wearing different masks. The lesson: **when several subsystems fail at once, look for a shared dependency (DNS, identity, time) before treating them as separate incidents.**

2. Timeline

When	Event
8/1 & 8/4	wlms.exe initiates DC shutdown (System Event 1074) — recurring, ~hourly once eval expired
8/4 ~9:02 PM	DC down during a window → NSX local credentials (audit/admin/root) marked Disconnected in SDDC Manager
8/5 (day)	User reports: VCF Ops Tasks "Service or view is not available"; Aria offline; License banner "licenses no longer associated"
8/5	Diagnosed licensing/Fleet failure → traced to vCenter adapter "Unable to connect to VC" → traced to DNS → DC found powered off
8/5	DC root cause identified: Server 2019 Eval expired (wlms); re-armed → DNS restored → adapters self-healed → licensing green
8/5	4 stuck OPERATIONS_MANAGER locks cleared via platform-DB delete; esxi04 service account recreated; vCenter root set to never-expire

3. The Cascade Map

DC (Server 2019 Eval) expires → wlms.exe shuts the DC down ~hourly → **DNS server gone** →

- VCF Ops can't resolve vcenter.lab.local → **vCenter + vSAN adapters "Unable to connect to VC"** → **license sync fails** → **Fleet Management (Passwords/Tasks) unavailable** → **vCenter root password change blocked**
- (pre-existing) failed esxi04 remediate holds **4 resource locks** → every credential op blocked
- (pre-existing) esxi04 **service account missing** on the host → remediate can never succeed

4. Layer-by-Layer RCA & Verified Fix

4.1 Root cause — DC Windows Server Evaluation expired

Root cause. The DC WINDOWS (windows.lab.local, 192.168.1.230) runs **Windows Server 2019 Standard Evaluation** (TIMEBASED_EVAL). When the eval expires (LicenseStatus=Notification , GracePeriodRemaining=0), the **Windows License Monitoring Service (wlms.exe)** force-shuts the machine down ~hourly — System log **Event 1074**, process C:\Windows\system32\wlms\wlms.exe . The DC is **not** in the nested vCenter inventory, so vCenter cannot see or restart it, which is why the drops looked random.

Fix (verified). On the DC, elevated: `slmgr.vbs /rearm` then reboot. This resets the 180-day evaluation clock (**6 re-arms remained** $\approx$ ~3 years of runway). Permanent alternative: convert Eval $\rightarrow$ full with a genuine Server 2019 Standard key via `DISM /online /Set-Edition:ServerStandard /ProductKey:<key> /AcceptEula` Or `slmgr /i pk <key> + slmgr /ato`. Diagnose with `slmgr /dlv` (re-arm count) and `slmgr /xpr` (expiry).

4.2 DNS loss (the propagation layer)

Symptom. `nslookup vcenter.lab.local 192.168.1.230` timed out; TCP/53 refused; DC ping 100% loss. Yet vCenter itself was reachable by IP (HTTP 200), which is the tell that this was **name resolution, not the target**.

Fix. Restoring the DC (4.1) restored DNS immediately. **Hardening (recommended, not yet done):** add `/etc/hosts` fallback entries for `vcenter.lab.local` (and the other management FQDNs) on the **VCF Operations appliance** so a DC/DNS blip cannot take licensing and Fleet Management down again.

4.3 VCF Operations vCenter / vSAN adapters — "Unable to connect to VC"

Symptom. In VCF Operations the `VMWARE` (`vcenter.lab.local`) adapter reported **"Unable to connect to VC"** and the `VirtualAndPhysicalSANAdapter` reported **"Cannot parse vSAN SDK version."** License sync rides on the vCenter adapter, so licensing failed downstream (KB 429366 / KB 421016).

Fix (verified). No manual action was required once DNS returned — the adapters **self-healed** on the next collection cycle to `DATA_RECEIVING / STARTED`. Confirmed via `GET /suite-api/api/resources?adapterKind=VMWARE`. *If they had not recovered:* Administration $\rightarrow$ Integrations $\rightarrow$ Accounts $\rightarrow$ `vcenter.lab.local` $\rightarrow$ Edit $\rightarrow$ Validate Connection (accept new cert if prompted; re-enter credentials if rejected), then re-trust the vCenter VMCA root under Administration $\rightarrow$ Control Panel $\rightarrow$ Trusted Certificates.

4.4 License sync failure & Fleet Management outage

Symptom. License Management: red **"Licenses could not be synchronized with the vCenter systems"** and **"Managed by VCF Instance: --"**; Fleet Management Tasks: **"Service or view is not available at this time."** The licenses themselves were **valid** (vCenter "Fully Licensed: Yes", entitlement to Oct 12 2026) — this was a **sync/adaptor** failure, not an expiry.

Fix (verified). Once the vCenter adapter was collecting again, re-running **Assign Primary License** (VCF cores) + **Assign Add-on** (vSAN) succeeded and the red banner cleared. Fleet Management (Tasks / Passwords) returned. VCF Operations cluster health was confirmed healthy throughout via CASA (`cluster_state INITIALIZED`, `online_state ONLINE`).

Related gate discovered. The **VCF Instances** password-management path (where the vCenter *root* account lives) shows "**not applicable for your license.**" Root cause (KB 406580): the vCenter cloud account is added for **monitoring only** because its account lacks the `Global.Licenses` privilege — the documented fix is to activate **system-managed credentials** (Integrations → VCF account → : → Edit → Manage Integrations). In this environment License + plug-in management were already active, so the vCenter root change was handled the alternative way (see 4.7).

4.5 Four stuck SDDC Manager resource locks

Root cause. A prior **failed "Credentials remediate operation"** for **esxi04 / svc-vcf-esxi01** acquired **4 resource locks** (`domain`, `system`, `nsx`, `vcenter`, all one `operationId`, `serviceId = OPERATIONS_MANAGER`) and died without releasing them. Every subsequent credential/lifecycle operation then failed at "Acquire Lock."

Fix (verified). The REST API **cannot** release locks on 9.0.1 (`DELETE /v1/resource-locks/{id}` → HTTP 400 "method not supported"). The **KB 432285 service restart** (`sddcmanager_restart_services.sh`) also did **not** clear these particular locks. The working fix was the **platform-DB delete** (RUNBOOK-SDDC-Manager-Resource-Lock-DB-Cleanup): open a temporary `trust` auth window in `pg_hba.conf`, `DELETE FROM lock;` (removed 4 rows), verify 0, then **restore** `pg_hba.conf`. Gate: 0 in-progress tasks. See the dedicated runbook **RUNBOOK-SDDC-Manager-Resource-Lock-API-Inspection-and-Release-2026-08-05**.

4.6 Missing ESXi service account on esxi04

Root cause (KB 416351). `esxi04` was rebuilt / re-added without the SDDC decommission-commission cycle, so the service account SDDC expects (`svc-vcf-esxi01`) was **absent from the host** (the host carried a differently-named `svc-vcf-esxi04` instead). Because "Remediate Password is non-functional until the account exists," every remediate **failed and jammed the 4 locks** from 4.5.

Fix (verified, KB 314643). Retrieved the stored password from SDDC (`GET /v1/system/credentials/service`, correlating host UUID → hostname so the right secret was used), then on `esxi04`: `esxcli system account add --id svc-vcf-esxi01 --password '<stored>' --password-confirmation '<stored>'` and `esxcli system permission set --id svc-vcf-esxi01 --role Admin`. Result: the account now exists with the correct password, and **remediate now fails cleanly (0 locks) instead of jamming** — the dangerous behavior is resolved.

Residual (open, low-impact). The credential's UI "Active" badge still will not flip: remediate fails prevalidation with `PASSWORD_MANAGER_RESOURCES_UNAVAILABLE` — `esxi04` not in `ACTIVE` state. The host's DB `status` was corrected `ERROR` → `ACTIVE` (verified) and SDDC services were restarted, yet prevalidation still fails — indicating an internal resource-state beyond `host.status`. **esxi04 is fully operational** (`CONNECTED` / `ASSIGNED` / `running`); this is a **Broadcom support item** for the stuck state, not an operational fault. Do **not** decommission/re-commission for a cosmetic badge.

4.7 vCenter root password expiry

Symptom. Health check: "vCenter root password expires in 6 days." The VCF-managed change path was gated by the licensing feature limitation (4.4).

Fix (verified). Set root to non-expiring in the vCenter VAMI — `https://vcenter.lab.local:5480` → Administration → Password expiration settings → Edit → **Password expires: No** → Save. No password value change → no SDDC desync → no licensing gate. (On production, prefer a long max-days over never-expire for compliance.)

5. Preventive Actions

#	Action	Why
1	Convert the DC off Evaluation (real Server key) — or diarise the re-arm	Eliminates the recurring <code>wlms</code> shutdown at the source
2	Stand up a second DNS resolver (or add <code>/etc/hosts</code> fallbacks on the VCF Ops appliance for the management FQDNs)	A single DC = single point of failure for the entire fleet's name resolution
3	Never run Remediate against a missing/mismatched account — recreate the account first (KB 314643)	A failed remediate jams 4 resource locks that then require platform-DB surgery
4	Treat simultaneous multi-subsystem failures as a shared-dependency problem first (DNS / identity / time)	Four "product bugs" here were one DNS outage
5	Keep the DB-cleanup runbook handy; the API cannot release locks on 9.0.1	Fastest safe path back from a jammed <code>OPERATIONS_MANAGER</code> workflow

6. Open Items

Item	Status / Next step
esxi04 credential "Active" badge	Broadcom support case for stuck <code>PASSWORD_MANAGER_RESOURCES_UNAVAILABLE</code> state; host is fully functional meanwhile

3 NSX local creds (audit/admin/root) "Disconnected"	Stale status from the 8/4 outage; NSX healthy (cluster ACTIVE, service account Active). Remediate via SDDC Password Management UI when convenient
vcf-ops root "Disconnected"	Fleet-managed (not in SDDC API): VCF Ops → Fleet Management → Passwords → VCF Management → vcf-ops root → Remediate
DNS resiliency	Add secondary resolver / hosts-file fallback (Preventive Action #2)

7. Reference

Source	Topic
KB 416351 / KB 314643	ESXi service account disconnected / re-create missing service account
KB 432285	Password remediation fails "Unable to acquire resource level lock(s)" — service restart
KB 441253	Password remediation "Resources not in ACTIVE state" — platform-DB status
KB 406580 / KB 436471	vCenter not eligible for management / License management — <code>Global.Licenses</code> & system-managed credentials
KB 429366 / KB 421016	License sync failure / VCF Ops vCenter adapter cert re-trust
KB 327195	Retrieve service-account credentials from SDDC Manager
VC LLC runbooks	RUNBOOK-SDDC-Manager-Resource-Lock-DB-Cleanup; RUNBOOK-SDDC-Manager-Resource-Lock-API-Inspection-and-Release-2026-08-05

8. Revision History

Version	Date	Author	Change
1.0	August 5, 2026	Virtual Control LLC	Initial release. Full multi-layer RCA of the 8/4–8/5 cascade from the DC evaluation-license expiry through DNS, VCF Operations adapters, licensing, Fleet Management, resource locks, and the esxi04 service account, with verified fixes and preventive actions.